Suppressed Density Fluctuations in a Disk-Sensing Brownian Model

Notes on the model of Wu et al.

1 The particle model and a stable suppression regime

Wu et al. [1] consider Brownian displacements whose local variance is controlled by the number of particles in a sensing disk. In the paper's dimensionless units, the perceived density and diffusivity are

$$P_i = \frac{n_i}{\pi R^2}, \qquad D(P_i) = \chi + (P_i - 1)^2, \qquad \chi = 0.002, \tag{1}$$

where the disk count n_i includes particle i itself. Our particle simulations use the same Itô Euler–Maruyama update, $R = 5$, $\Delta t = 0.05$, and periodic squares of side $L = 40$ or 80. Following Ref. [1], the mean density $\rho_0 = N/L^2$ takes the values $1.6, 2, 3$.

For a homogeneous reference state, replace the particle count by the normalized disk convolution $P = W_R * \rho$, where $W_R(r) = \Theta(R - r)/(\pi R^2)$. Linearizing the Itô–Dean density equation about $\rho = \rho_0$ gives

$$\bar{D} = \chi + (\rho_0 - 1)^2, \quad g = D'(\rho_0) = 2(\rho_0 - 1), \quad \widehat{W}_R(k) = \frac{2J_1(kR)}{kR}, \quad S_{\mathrm{lin}}(k) = \frac{\bar{D}}{\bar{D} + \rho_0 g \widehat{W}_R(k)}. \tag{2}$$

Here $\bar{D} = D(\rho_0)$ is the diffusivity at the mean density, and g is its local sensitivity to perceived density. The factor $\widehat{W}_R(k)$ is the Fourier transform of the normalized sensing disk at wave number k; J_1 is the first-order Bessel function. Finally, $S_{\mathrm{lin}}(k)$ is the predicted structure factor, normalized so an uncorrelated state has $S = 1$. It follows from the stationary covariance of each conserved Gaussian Fourier mode and requires $\bar{D} + \rho_0 g \widehat{W}_R(k) > 0$ for all allowed modes. This is a reference for a stable homogeneous state, not an exact spectrum of the nonlinear particle process. The continuum replacement also neglects the tagged-particle self-count shift $1/(\pi R^2)$.

For $\rho_0 > 1$, $g > 0$ and $S_{\mathrm{lin}}(0^+) = 1/(1 + B) < 1$, where $B = \rho_0 g/\bar{D}$. The negative minimum of the disk transform is approximately -0.13228, so the continuum homogeneous state remains linearly stable throughout the resolved band if $B < 7.56$. Here $B \simeq 5.30, 3.99, 3.00$ for $\rho_0 = 1.6, 2, 3$, respectively. These are the conditions under which suppression relative to the uncorrelated baseline $S = 1$ can be assessed without first crossing a linear instability.

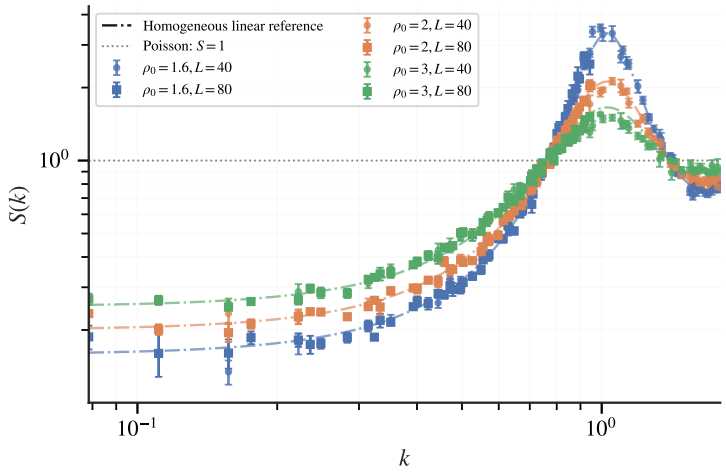


Figure 1: Measured structure factor versus k (points). Dot–dash curves are the homogeneous linear reference, Eq. (2); circles and squares denote $L = 40$ and 80. The dotted line marks the uncorrelated baseline $S = 1$.

2 Numerical evidence and its limits

Figure 1 shows a marked drop below $S = 1$ as k decreases through the sensing-scale crossover. Fits to the saved $L = 80$ spectra over the finite interval $0.5 \le k \le 1$ give effective log–log slopes of about $3.61, 2.76, 1.89$ for $\rho_0 = 1.6, 2, 3$; they are not asymptotic exponents. At the smaller modes supplied by $L = 80$, $k_{\min}R = 2\pi R/L \simeq 0.393$ and the measured first-shell values are

$$S(k_{\min}) = 0.187 \pm 0.021 \quad (\rho_0 = 1.6), \qquad 0.235 \pm 0.045 \quad (\rho_0 = 2), \qquad 0.268 \pm 0.014 \quad (\rho_0 = 3). \tag{3}$$

Errors are standard errors from six time blocks over $500 < t \le 1500$ in one trajectory per parameter set; they are not independent-seed uncertainties. The $L = 40$ spectra extend to larger k and agree broadly with $L = 80$ in their overlap.

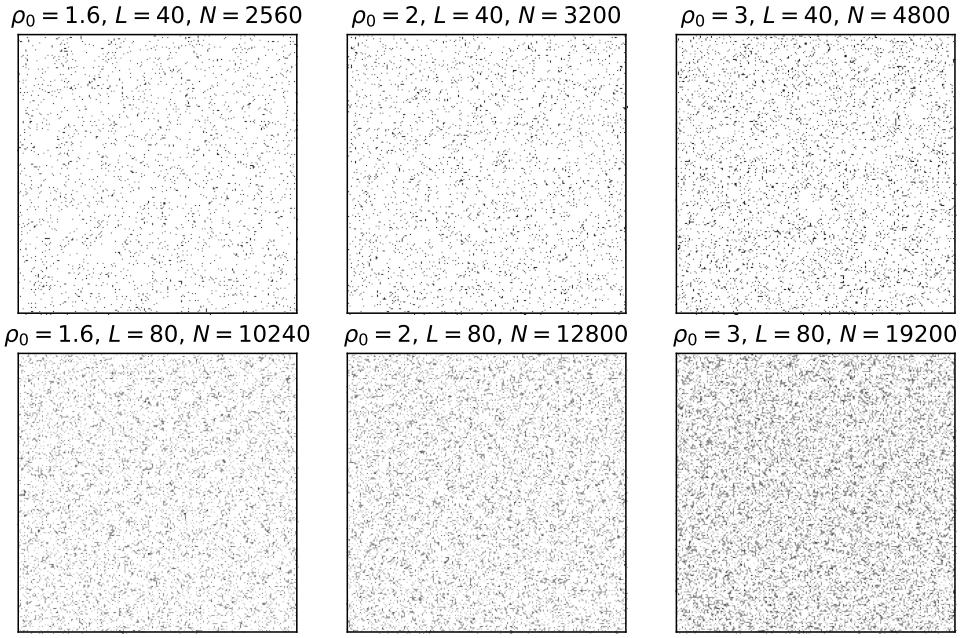


Figure 2: Final particle positions at $t = 1500$ for the six runs in Fig. 1: columns $\rho_0 = 1.6, 2, 3$; rows $L = 40, 80$. Each panel displays the whole periodic square. The suppression claim comes from $S(k)$, not visual appearance.

Conclusion

The equal-weight disk-sensing model of Ref. [1] *can suppress density fluctuations* over measured long wavelengths when $\rho_0 > 1$, the positive response g is strong enough to reduce S, and the homogeneous state remains stable. This does not require inverse-distance sensing. Finite R and finite B give a positive $k \to 0$ plateau in the linear theory; the finite-interval slopes in Fig. 1 therefore do not establish asymptotic hyperuniformity.

[1] B. Wu et al., *Nonlocal Sensing Drives Hybrid Phase Separation in Brownian Matter*, arXiv:2606.21464v1 (2026), especially the particle protocol in Supplemental Material S1.